

4 hours on examples

Problem 1 (13PUMACFA3)

FTSOC, let there be such a G with more than $2(k-1)^2$ edges. There are no k -stars if and only if the degree of every vertex is less than k .

We prove there is a k -matching. We select edges from the grid one at a time. Once we select an edge, we disable itself and all other edges connected the vertices at the ends of the edge. Since the degree of every vertex is at most $k-1$, we've disabled at most $2(k-1)$ edges.

There are more than $2(k-1)^2$ edges in total, so after $(k-1)$ choices, there will still be at least one more edge unchosen. We choose another such edge to have k edges chosen in total. Clearly, no two of these share a common endpoint, so we are done.

Problem 3 (03IMO1)

Let $a_1, a_2, \dots, a_{101}$ be the numbers in A . WLOG, sort A such that $a_1 < a_2 < \dots < a_{101}$.

We construct $t_1, \dots, t_{100}$ one at a time. We let X be the set of "open" numbers at any point, which is the set of numbers we may set the next t_i to such that A_i continues being pairwise distinct with $A_1 \dots A_{i-1}$.

When we set t_i , we remove numbers from X that we can no longer set t_{i+1} as without having A_{i+1} share terms with A_i . These numbers would be, [citation needed?]

$$\begin{aligned} & (t_i + a_1) - a_1, (t_i + a_1) - a_2, \dots, (t_i + a_1) - a_{101}, \\ & (t_i + a_2) - a_1, (t_i + a_2) - a_2, \dots, (t_i + a_2) - a_{101}, \\ & \dots \\ & (t_i + a_{101}) - a_1, (t_i + a_{101}) - a_2, \dots, (t_i + a_{101}) - a_{101}, \end{aligned}$$

In the beginning, we may set t_1 to anything so $X = \{1, \dots, 10^6\}$. At any step, we choose the next t_i to be the smallest element in X .

Since $t_i + a_i - a_j < t_i$ if $j > i$, of the 101^2 numbers we would need to remove from X , the $\frac{100(101)}{2}$ numbers,

$$\begin{aligned} & (t_i + a_1) - a_2, \dots, (t_i + a_1) - a_{101}, \\ & (t_i + a_2) - a_3, \dots, (t_i + a_1) - a_{101}, \\ & \dots, \\ & (t_i + a_{100}) - a_{101} \end{aligned}$$

Are smaller than the smallest number currently in X , so they already aren't present in X . Therefore, at any step, we'd remove at most $101^2 - \frac{100(101)}{2} < 10000$ numbers from X . Since there were 10^6 numbers to begin with, we'd be able to do this at least 100 times.

5 clubs; 5.5 hours

Problem 6 (10MPO4)

The construction is as follows, at any time, we look at how many points in S are collinear in each vertical or horizontal line. For any of the lines (possibly tied) with most points on it, we group those into the same subset and remove it from S .

We call the subsets we form $A_1, A_2, \dots, A_m$. We now go in reverse adding the points back subset by subset. We define $S_k = A_1 \cup A_2 \cup \dots \cup A_k$. Note that $S_m = S$. We claim that by induction, S_k obeys the problem statement if we select $A_1 \dots A_k$ to be the subsets.

Base case: $k = 1$

All the points in the set are in the same subset. Therefore, there are zero pairs of distinct collinear points in different subsets.

Inductive step:

Let's say the new subset adds x collinear points line l to the previous y we had. Clearly, none of the previous y points may also lie l for otherwise we would have grabbed it when generating the subsets.

Note that each of the x points may be collinear with at most $x - 1$ other points of the previous y we had. If there was any more, then we would have at least $x + 1$ points in a line, so we would have picked this line instead of l .

Therefore, we have added at most $x(x - 1)$ new intersections each with distinct points out of y . We now prove that,

$$(y + x)^{\frac{3}{2}} - y^{\frac{3}{2}} \geq x(x - 1)$$

Note that since $y \geq x(x - 1)$,

$$\begin{aligned} &\Leftrightarrow (x(x - 1) + x)^{\frac{3}{2}} - (x(x - 1))^{\frac{3}{2}} \geq x(x - 1) \\ &\Leftrightarrow x^3 - x(x - 1) \geq (x(x - 1))^{\frac{3}{2}} \\ &\Leftrightarrow (x^3 - x^2 + x)^2 \geq (x^2 - x)^3 \\ &\Leftrightarrow x^6 - 2x^5 + 3x^4 - 2x^3 + x^2 \geq x^6 - 3x^5 + 3x^4 - x^3 \\ &\Leftrightarrow x^5 \geq x^3 - x^2 \end{aligned}$$

This is clearly true.

Conclusion: By induction we are done.

Problem 7 (04RUS106)

We start with any city.

Claim 1: Each city can reach 167 other cities in the republic through the republic.

Since there are only 333 cities outside the republic, the 500 roads leading out of the city can reach at least 167 other cities.

Claim 2: Our city can reach at least 334 other cities in the republic through the republic.

For the sake of contradiction, may this city only reach $x < 334$ other cities. Let the $x + 1$ cities here that cannot reach any other cities in the republic be “isolated”. Each isolated city must reach at least 167 other cities in the republic which must be “isolated”. Therefore, there are $(x + 1) \cdot 167 > \binom{x + 1}{2}$ distinct edges between these isolated cities. (!)

Claim 3: Our city can reach all other cities in the republic through the republic.

We do this inductively, proving that our city can reach n other cities in the republic for all $667 \geq n \geq 334$.

Base case: $n = 334$, done with claim 2.

Inductive step:

Let's say we have proven that the city may reach $k - 1$ other cities presently. We show that in fact it may reach k cities.

For the sake of contradiction, let k other cities be impossible. We call the k cities (the starting city as well as the $k - 1$ it can reach) “isolated”. All the paths leading out of each of the “isolated” cities must either lead to other isolated cities or to the 333 cities outside the republic. There are $500k$ such roads, of which $\binom{k}{2}$ at most may run between “isolated cities”. The other $500k - \binom{k}{2}$ must run to the cities outside the republic.

In total, there are $500 \cdot 333$ roads that end at a city outside of the republic. Note that the 333 cities outside the republic have $\binom{333}{2}$ paths between them so we already have $166 \cdot 333$ here. Therefore, the 333 cities outside the republic accepts a total of $334 \cdot 333$ roads.

$$500k - \binom{k}{2} > 334 \cdot 333$$

This is impossible on $334 < k < 667$.

Conclusion: By mathematical induction our city must reach at least 667 other cities.

We are hence done.

14 clubs; 7.5 hours

Problem 8 (19SLC2)

We prove an even stronger claim, that given any n blocks with weight at least 1 and any total weight k as long as $k \leq 2n$, for any real number $r \in [0, k - 2]$, there is a subset of the blocks whose total weight is between $r, r + 2$ inclusive. We do this by induction on n .

Base case: $n = 1$

The weight of the only weight is the total weight, which is between 1, 2. Both works.

Inductive step:

Let the total weight be $w_{tot} \leq 2n + 2$.

For any set of $n + 1$ blocks with weight $k \leq 2n + 2$, we take out the heaviest block, which clearly must have weight $w_{tot} - n \geq w \geq 2$.

With the remaining n blocks with weight w_n totaling to at least n , by the inductive hypothesis, for any real number $r \in [0, w_n - 2]$ there is a subset of blocks with total weight between r and $r + 2$ inclusive.

For $r \in [0 + w, w_n + w - 2]$, we may take these constructions and add our heaviest block of weight w . Note that $w_n + w = w_{tot}$.

For $r \in [w_n - 2, w_n]$, there is a subset with weight w_n using the remaining n blocks.

For $r \in [w - 2, w]$, the construction would be using the heaviest weight alone.

Note that $[0, w_n - 2] \cup [w_n - 2, w_n] \cup [w - 2, w] \cup [w, w_{tot} - 2] = [0, w_{tot} - 2]$ since $w_n \geq w - 2$.

Conclusion: Using mathematical induction, we are done.

Problem 10 (14IMO5)

We prove that this is true where the total value is at most $k - \frac{k}{2k+1}$ and we may construct at most k collections.

First, if there are any pairs of coins with size $\frac{1}{2n}$, we force these two to be in the same collection so effectively we may remove them and replace it with a coin of size $\frac{1}{n}$. We do this until no such pairs remain. After this process, there may only be at most one of each coin with size $\frac{1}{n}$ if n is even.

Next, if there is any group of n coins with size $\frac{1}{n}$, we force those to be in a same group of size 1. We do this until no more such groups remain. This way, for any odd number $2n + 1$, we additionally have there are at most $2n$ coins of $\frac{1}{2n+1}$ value.

We now form the k groups $G_1, \dots, G_k$.

G_1 will include the coins of size $\frac{1}{2}$. There may be at most 1 of these so the current size is less than 1.

G_i for $i > 1$ will include the coins of size $\frac{1}{2i-1}$ and $\frac{1}{2i}$. Note that there may be at most $2i - 2$ of the coins of size $\frac{1}{2i-1}$ and at most one $\frac{1}{2i}$ and $\frac{2i-2}{2i-1} + \frac{1}{2i} < \frac{2i-2}{2i-1} + \frac{1}{2i-1} = 1$.

Next, we just arbitrarily put the remaining coins with size at most $\frac{1}{2k+1}$ into any group that fit them. For the sake of contradiction, if this is impossible, then all the groups would be at size at least $1 - \frac{1}{2k+1}$ presently which means the total weight is already at least $k - \frac{k}{2k+1}$, the limit. Therefore, it is impossible to have any more unsorted coin.

26 clubs; 11.5 hours

Problem 11 (22AMO2)

We may draw a line through the center of the $2n$ -gon and one of the vertices. This divides the sides of the $2n$ -gon into two halves. Note that parallel sides have same color so one half is the reflection of the other over the center. Therefore, we claim that all colorings of the first half with b black rods and w white rods will result in possible polygons B, W .

We let each rod of the $2n$ -gon be vectors $v_1, v_2, \dots, v_{2n}$ pointing clockwise, opposite vectors cancel out and the sum of black and white rods vectors is zero, meaning B, W can be polygons always.

We now show that going from one coloring to another does not affect the difference of area between W and B . To do this, we show the following.

Claim: Swapping two adjacent sides in the first half (and hence swapping the corresponding opposite sides on the second half) does not affect the difference in areas of the resulting polygons.

We define the first pair of B, W to be consisted of the $2b$ vectors, in clockwise order,

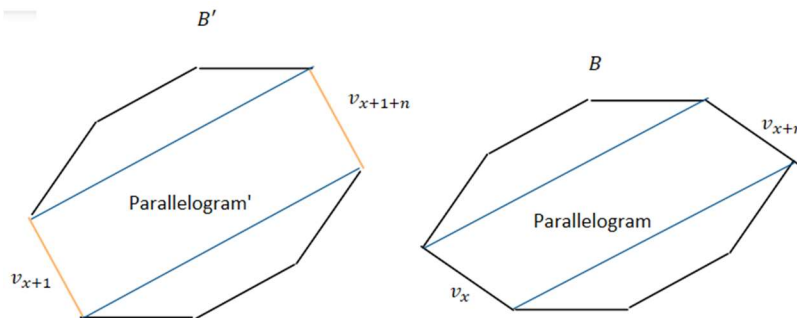
$$B: v_{a_1}, v_{a_2}, \dots, v_{a_i}, v_x, v_{a_{i+2}}, v_{a_{i+3}}, \dots, v_{a_b}, v_{a_1+n}, v_{a_2+n}, \dots, v_{a_i+n}, v_{x+n}, v_{a_{i+2}+n}, v_{a_{i+3}+n}, \dots, v_{a_b+n}$$

$$W: v_{z_1}, v_{z_2}, \dots, v_{z_j}, v_{x+1}, v_{z_{j+2}}, \dots, v_{a_w}, v_{z_1+n}, v_{z_2+n}, \dots, v_{z_j+n}, v_{x+1+n}, v_{z_{j+2}+n}, \dots, v_{a_w+n}$$

Where indices are taken modulo $2n$. After swapping v_x with v_{x+1} and v_{x+n} with v_{x+1+n} ,

$$B': v_{a_1}, v_{a_2}, \dots, v_{a_i}, v_{x+1}, v_{a_{i+2}}, v_{a_{i+3}}, \dots, v_{a_b}, v_{a_1+n}, v_{a_2+n}, \dots, v_{a_i+n}, v_{x+1+n}, v_{a_{i+2}+n}, v_{a_{i+3}+n}, \dots, v_{a_b+n}$$

$$W': v_{z_1}, v_{z_2}, \dots, v_{z_j}, v_x, v_{z_{j+2}}, \dots, v_{a_w}, v_{z_1+n}, v_{z_2+n}, \dots, v_{z_j+n}, v_{x+n}, v_{z_{j+2}+n}, \dots, v_{a_w+n}$$



We may draw vector $-v_{a_1} - v_{a_2} - \dots - v_{a_i}$ starting at the end of v_{a_i} and do the analogous for $v_{a_{i+n}}$ as well as on W . We may then also do this for B' and W' as well. For example, see the diagram to the right.

Note that the polygon formed by the vectors,

$$v_{a_1}, v_{a_2}, \dots, v_{a_i}, -v_{a_1} - v_{a_2} - \dots - v_{a_i} \\ v_{a_{n+1}}, v_{a_{n+2}}, \dots, v_{a_{n+i}}, -v_{a_{n+1}} - v_{a_{n+2}} - \dots - v_{a_{n+i}}$$

Stay the same in both B and B' . The only difference in area in these two polygons is the difference in area between parallelograms,

$$|B| - |B'| = |\text{Parallelogram}| - |\text{Parallelogram}'|$$

Using cross product,

$$= |v_x \times (-v_{a_1} - v_{a_2} - \dots - v_{a_i})| - |v_{x+1} \times (-v_{a_1} - v_{a_2} - \dots - v_{a_i})|$$

WLOG, let $x = 1$ (I should have done this earlier).

$$= |v_1 \times (v_{a_1} + v_{a_2} + \dots + v_{a_i})| - |v_2 \times (v_{a_1} + v_{a_2} + \dots + v_{a_i})|$$

We let $\vec{i} = v_{a_1} + v_{a_2} + \dots + v_{a_i}; \vec{j} = v_{z_1} + v_{z_2} + \dots + v_{z_j}$ (not unit vectors!). Therefore,

$$\begin{aligned} |B| - |W| &= |B'| - |W'| \\ \Leftrightarrow |B| - |B'| &= |W| - |W'| \\ \Leftrightarrow v_1 \times \vec{i} - v_2 \times \vec{i} &= v_1 \times \vec{j} - v_2 \times \vec{j} \\ \Leftrightarrow (v_1 - v_2) \times (\vec{i} + \vec{j}) &= 0 \end{aligned}$$

However, note that $\vec{i} + \vec{j} = v_3 + v_4 + \dots + v_n$.

$$\Leftrightarrow v_1 \times v_3 + v_1 \times v_4 + \dots + v_1 \times v_n = v_2 \times v_3 + v_2 \times v_4 + \dots + v_2 \times v_n$$

WLOG, let the side length of the $2n$ -gon be one.

$$\begin{aligned} \Leftrightarrow \sin\left(2 \cdot \frac{2\pi}{2n}\right) + \sin\left(3 \cdot \frac{2\pi}{2n}\right) + \dots + \sin\left((n-1) \cdot \frac{2\pi}{2n}\right) \\ = \sin\left(1 \cdot \frac{2\pi}{2n}\right) + \sin\left(2 \cdot \frac{2\pi}{2n}\right) + \dots + \sin\left((n-2) \cdot \frac{2\pi}{2n}\right) \end{aligned}$$

Note that $\sin\left(k \times \frac{2\pi}{2n}\right) = \sin\left(\frac{k}{n} \cdot \pi\right) = \sin\left(\frac{k-3}{n} \cdot \pi\right) = \sin\left((k-3) \times \frac{2\pi}{2n}\right)$, so this is true.

With the claim proven, we may then perform enough of these swaps until the desired configuration has been reached with no change in difference of area.

Problem 13 (18HMMTT6)

We build the subset $X \subset S$ one element at a time. Define $f(X)$ as the set of distinct residues modulo n subsets of X may currently sum to.

For our first element, we take any element $x \in S$ such that $n \nmid x$. With this, $|f(X)| = 2$ currently. We either have 0 by taking the empty subset or $n \bmod x$.

While subsets of X do not sum to every possible residue modulo n , we add other numbers in S we haven't yet chosen. We make sure to take only elements that increase $|f(X)|$. For the sake of contradiction, if this wasn't possible, then we consecutively add elements to X from S . Note that $f(X)$ won't change at all at any step. In the end, $f(X) = f(S) \neq \{0, 1, \dots, n-1\}$. (!)

We will need to add at most $n-2$ other elements before $|f(X)| = n$ and X is *comprehensive*. In total, we have $|X| \leq n-1$ as desired.

36 clubs; 13.5 hours

Problem 15 (17USATST1)

We claim the answer is $\left\lceil \frac{n}{t} \right\rceil$.

To prove this is maximal, consider the $\left\lceil \frac{n}{t} \right\rceil$ teams where no pair of teams share any signature colors. Since there are $\left\lceil \frac{n}{t} \right\rceil$ teams in total, that's our limit here.

To prove this is always attainable, we count how many teams use each of the remaining signature colors and take the least used color C . We add one of the teams who use this signature color to S .

We now eliminate this team as well as any team who was using this signature color. Obviously, no remaining teams will use signature color C anymore. Since no other signature color previously had less users than C previously, the only way for another color to no longer be used by any remaining team is if all the teams we had eliminated chose it. Obviously, at most t colors may be eliminated this way.

Repeatedly doing this, we may form at least $\left\lceil \frac{n}{t} \right\rceil$ teams.

Problem 17 (23CAN5)

We claim $\left\lceil \frac{n}{4k} \right\rceil$ is maximum.

One construction is to form $\left\lceil \frac{n}{4k} \right\rceil$ cliques of $4k$ or $4k-1$ members. If we were to partition each clique into two sets, there is at most $\left(\frac{4k}{2}\right)^2 = 4k^2$ edges between the cliques of $4k$ and at most $(2k)(2k-1) = 4k^2 - 2k$ edges between the cliques of $4k-1$ members. In total, since,

$$4k^2 \leq k(4k)$$

$$4k^2 - 2k \leq k(4k-1)$$

The number of edges between the two sets is less than k times the total number of vertices, and this satisfies the kn condition. Since we have $\left\lceil \frac{n}{4k} \right\rceil$ cliques, we cannot find an independent set of more than $\left\lceil \frac{n}{4k} \right\rceil$ vertices.

We now prove that we can always find an independent set of $\left\lceil \frac{n}{4k} \right\rceil$ vertices in any graph satisfying the conditions to finish. For the sake of contradiction, let this be impossible. We select the maximal independent set in our current graph, M_1 with $|M_1| < \frac{n}{4k}$. We remove the vertices of M_1 from the graph and all the edges connecting to it and find the next maximal independent set, M_2 . Note that $|M_2| \leq |M_1| < \frac{n}{4k}$ as well as the vertices of M_2 formed an independent set in the original graph. We do this until we have exactly $2k$ sets,

$$M_1, M_2, \dots, M_{2k}$$

Let S be the remaining vertices that were not in any of these sets. Since these sets were maximal, then for each M_i there must be at least one edge between one of its vertices and each element of S for otherwise we could have added that disconnected element to M_i . Since,

$$|S| = n - |M_1| - \dots - |M_{2k}| > n - \frac{n}{4k} \cdot 2k = \frac{n}{2}$$

In total, if we choose $M_1 \cup M_2 \cup \dots \cup M_{2k}$ and S as our two sets, we'd have,

$$2k \cdot |S| > kn$$

Edges between them. (!)

50 clubs; 15.5 hours

Problem 18 (16PTNMB3)

We take the largest triangle ABC with area x . We draw a line l_a through A parallel to BC , l_b through B parallel to CA , and l_c through C parallel to AB . We claim that all planes in the plane lie in or on the boundaries of this triangle. FTSOC, if this wasn't the case and there is such a point P that WLOG lies on the opposite side of l_a as BC , then $[PBC] > [ABC]$ (!).

Therefore, we dilate the final triangle about a point in its center such that its area increases to $4 > 4x$. All the points in the plane now lie strictly in its interior.

Problem 20 (ZC62C56E / Sylvester-Gallai)

FTSOC, let this not be possible and no line passes through 2 points. Let x be the minimum distance greater than zero between a point A and a line with points B, C, D on it in order. This exists by the extremal principal as there are finitely many points and such lines in the plane. WLOG, let C and D be on the same side of BCD as the perpendicular from A . Note that the distance from C to AD is less than x .

56 clubs; 16.5 hours